GI Neuroendocrine Tumors

Gastric Neuroendocrine Tumors

Category	Differentiation	Grade	Mitotic Rate	Ki-67 Index
NET - Grade 1	Well-diff	Low	<2	<3%
NET - Grade 2	Well-diff	Intermed	2-20	3-20%
NET - Grade 3	Well-diff	High	>20	>20%
NEC - Large Cell	Poorly	High	>20	>20%
NEC - Small Cell	Poorly	High	>20	>20%
MiNEN	Variable	Variable	Variable	Variable

Mixed Neuroendocrine - Nonneuroendocrine Neoplasms

Somatostatin Receptor (SSTR) PET for Staging

Reference Here

Frequently performed with ^{68}Ga-DOTATATE Most useful for poorly-differentiated or high-grade tumors Less helpful for small tumors

Gastric Neuroendocrine Tumors

Type 1 Multifocal Tumors - Driven by atrophic gastritis (70–80% of gastric NETs):

- Atrophic gastritis $\rightarrow \downarrow$ gastric acid ($\uparrow$pH) $\rightarrow \uparrow$ gastrin from G cells
- Gastrin is trophic for enterchromaffin-like (ECL) cells $\rightarrow$ Neuroendocrine tumors
- Small (<1 cm), multifocal, well-differentiated, indolent

Type 2 Multifocal Tumors - Driven by Zollinger-Ellison Syndrome (rare)

- Gastrinoma produces high levels of gastrin and low gastric pH
- Generally have accompanying ulcer disease

Type 3 Sporadic Tumors - Autonomous growth (normal pH and normal gastrin)

Highest metastatic potential; most aggressive subtype

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- Small (<1 cm), multifocal, well-differentiated, indolent
- Small tumors <1cm: Endoscopic resection
- Tumors 1-2cm: Endoscopic mucosal resection (EMR) or Endoscopic Submucosal Disssection (ESD)
- High risk tumors (>2cm, Node$^+$, Grade 3, muscle invasion): Surgical resection

Gastric Neuroendocrine Tumors

Type 2 Multifocal Tumors - Driven by Zollinger-Ellison Syndrome (rare)

- Gastrinoma produces high levels of gastrin and low gastric pH
- Generally have accompanying ulcer disease
- Localization of gastrinoma (and resection) critical to treatment

Gastric Neuroendocrine Tumors

Type 3 Sporadic Tumors - Autonomous growth (normal pH and normal gastrin)

Highest metastatic potential; most aggressive subtype

- Small tumors <1cm: Endoscopic resection
- High risk tumors: Surgical resection with node dissection

Mixed Neuroendocrine / Nonneuroendocrine Neoplasms

- Tumors showing neuroendocrine *and* nonneuroendocrine differentiation
- Tend to be aggressive with liver and nodal metastasis
- Origin from a single stem cell that then gives rise to two differentaition paths
- Small tumors treated with surgery
- Larger tumors treated with chemotherapy
- Also known as Mixed Adenoneuroendocrine Carcinoma (MANEC)

(**jacob721?**)

Lutathera Lu^{177}-dotatate

- Radioactive-tagged dotatate (artificial somatostatin analogue)
- Indicated for treatment of somatostatin recpetor (SSTR)-positive tumors
- Gastroenterohepatic neuroendocrine tumors of foregut, midgut, and hindgut
- Dosed IV every 8 weeks x 4 doses, folloed by long-acting octreoride every 4 weeks

NETTER-1 clinical trial

Eligibility criteria:

- Neuroendocrine tumors
- Ki-67 index $ $20%
- OctreoScan positive in all lesions
- Disease progression on standard-dose octroetide

Randomized:

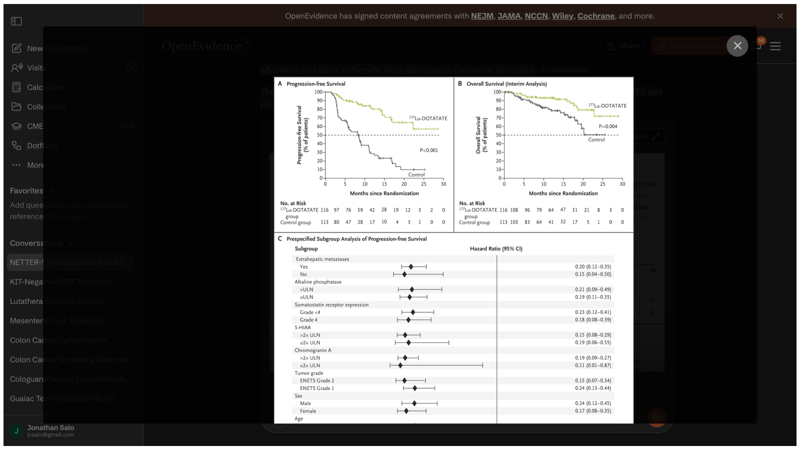
- ^{177}Lu-dotatate q8 weeks x4 + octreotide LAR 30mg IM OR
- Octreotide LAR 60mg IM every4 weeks

	^{177}Lu-dotatate	Control
Progression-free survival @ 20mo	65%	11%
Median PFS	(not reached)	8.4mo
Objective Response Rate	18%	3%
Overall Survival Median	48mo	33mo

Survival difference was likely diluted by crossovers at progression

More pronounced survival benefit for patients with KPS 60-80

NETTER-1



Lutathera indications for GI/Pancreatic tumors

- Well differentiated (Grade 1/2) tumors
  - Progression on octreodide LAR or lanreotide
  - Significant tumor burden and Ki-67 index $ $10% (first line)
- Well differentiated Grade 3 tumors
  - Favorable biology (Ki-67 10%-55%) - supported by NETTER-2 clinical trial

Netter-2 Trial - Grade 2/3 Neuroendocrine tumors

226 Patients with higher grade 2 (Ki67 10% and 20%) and grade 3 (Ki67 >20% and 55%), somatostatin receptor-positive gastroenteropancreatic neuroendocrine tumours

Treated with 4 cycles of Lutathera → Octreotide 30mg IM q4 wk

vs

Octreotide 60mg IM q4wk

Impact of Liver Tumour Burden, Alkaline Phosphatase Elevation, and Target Lesion Size on Treatment Outcomes With Lu-Dotatate: An Analysis of the NETTER-1 Study. European Journal of Nuclear Medicine and Molecular Imaging. 2020. Strosberg J, Kunz PL, Hendifar A, et al.’

Cost-Effectiveness of [177Lu]Lu-Dotatate for the Treatment of Newly Diagnosed Advanced Gastroenteropancreatic Neuroendocrine Tumors: An Analysis Based on Results of the NETTER-2 Trial. Journal of Nuclear Medicine : Official Publication, Society of Nuclear Medicine. 2025. Holzgreve A, Unterrainer LM, Tiling M, et al.

Progression-free survival with Lutathera was 22.8mo (vs 8.5mo)

⇒ Lutathera is indicated for first-line therapy of higer grade neuroendocrine tumors

(**singh2807?**)